LAB 04

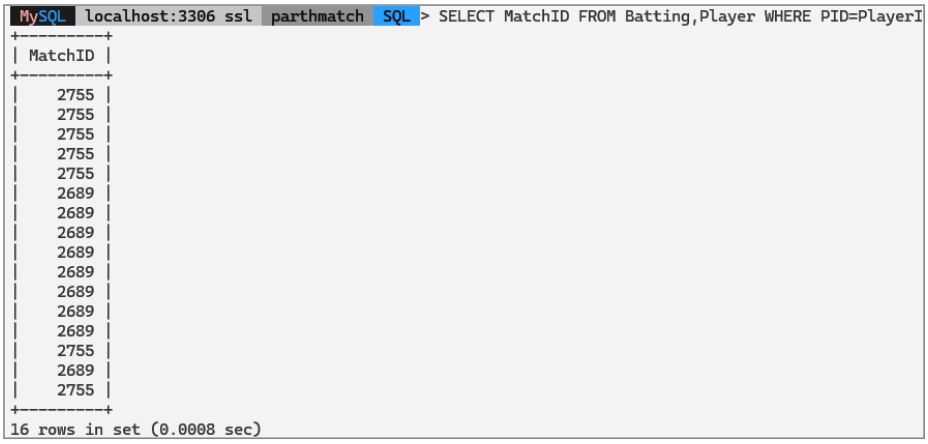
Aim : To implement and execute advanced SQL queries on a relational database using joins, subqueries, set operations, aggregate functions, grouping, and conditional operators for effective data retrieval and analysis.

Q1. Find match Ids of all the matches in the database in which tendulkar batted.

mysql> SELECT MatchID FROM Batting, Player;



mysql> SELECT MatchID FROM Batting,Player WHERE PID=PlayerID;



mysql> SELECT MatchID FROM Batting,Player WHERE PID=PlayerID AND Lname='Tendulkar';

OR

mysql> SELECT MatchID FROM Batting WHERE PID IN (SELECT PlayerID FROM Player WHERE LName='Tendulkar');

```
MySQL localhost:3306 ssl parthmatch SQL > SELECT MatchID FROM Batting,Player WHERE PID=PlayerID AND Lname='Tendulkar';
+---------+
| MatchID |
+---------+
|    2689 |
+---------+
1 row in set (0.0012 sec)
MySQL localhost:3306 ssl parthmatch SQL >
```

Q2. Find match information of all matches in which Dohni has batted.

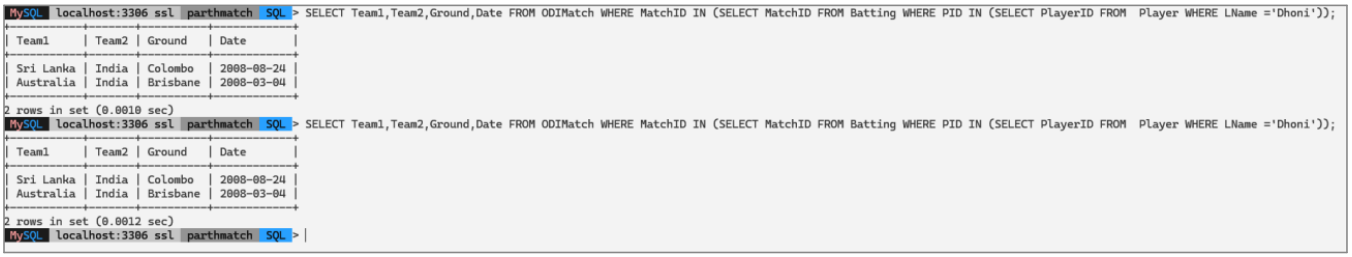
mysql> SELECT Team1,Team2,Ground,Date FROM ODIMatch,Batting,Player;

West Indies	India	Kingston	2006-05-18
West Indies	India	Kingston	2006-05-18
West Indies	India	Kingston	2006-05-18
West Indies	India	Kingston	2006-05-18
West Indies	India	Kingston	2006-05-18
West Indies	India	Kingston	2006-05-18
West Indies	India	Kingston	2006-05-18
West Indies	India	Kingston	2006-05-18
West Indies	India	Kingston	2006-05-18
West Indies	India	Kingston	2006-05-18
India	England	Delhi	2006-03-28
India	England	Delhi	2006-03-28
India	England	Delhi	2006-03-28
India	England	Delhi	2006-03-28
India	England	Delhi	2006-03-28
India	England	Delhi	2006-03-28
India	England	Delhi	2006-03-28
India	England	Delhi	2006-03-28
India	England	Delhi	2006-03-28
India	England	Delhi	2006-03-28
India	England	Delhi	2006-03-28
India	England	Delhi	2006-03-28
India	England	Delhi	2006-03-28
India	England	Delhi	2006-03-28
India	England	Delhi	2006-03-28
Pakistan	India	Rawalpindi	2006-02-11
Pakistan	India	Rawalpindi	2006-02-11
Pakistan	India	Rawalpindi	2006-02-11
Pakistan	India	Rawalpindi	2006-02-11
Pakistan	India	Rawalpindi	2006-02-11
Pakistan	India	Rawalpindi	2006-02-11
Pakistan	India	Rawalpindi	2006-02-11
Pakistan	India	Rawalpindi	2006-02-11
Pakistan	India	Rawalpindi	2006-02-11
Pakistan	India	Rawalpindi	2006-02-11
Pakistan	India	Rawalpindi	2006-02-11
Pakistan	India	Rawalpindi	2006-02-11
Pakistan	India	Rawalpindi	2006-02-11
Pakistan	India	Rawalpindi	2006-02-11
Pakistan	India	Rawalpindi	2006-02-11
Pakistan	India	Rawalpindi	2006-02-11
Pakistan	India	Peshwar	2006-02-06
Pakistan	India	Peshwar	2006-02-06
Pakistan	India	Peshwar	2006-02-06
Pakistan	India	Peshwar	2006-02-06
Pakistan	India	Peshwar	2006-02-06
Pakistan	India	Peshwar	2006-02-06
Pakistan	India	Peshwar	2006-02-06
Pakistan	India	Peshwar	2006-02-06
Pakistan	India	Peshwar	2006-02-06

mysql> SELECT Team1,Team2,Ground,Date FROM ODIMatch WHERE MatchID IN (SELECT MatchID FROM Batting WHERE PID IN (SELECT PlayerID FROM Player WHERE LName ='Dhoni'));

OR (Using Join)

mysql> SELECT Team1,Team2,Ground,Date FROM ODIMatch,Batting,Player WHERE ODIMatch.MatchID=Batting.MatchID AND PlayerID=PID AND LName='Dhoni';

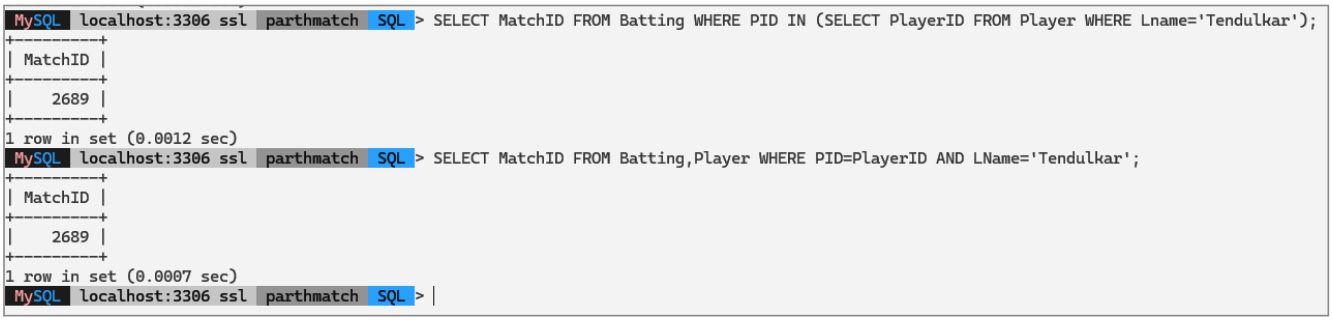


Q3. Find match Ids of all matches in which Tendulkar batted.

mysql> SELECT MatchID FROM Batting WHERE PID IN (SELECT PlayerID FROM Player WHERE Lname='Tendulkar');

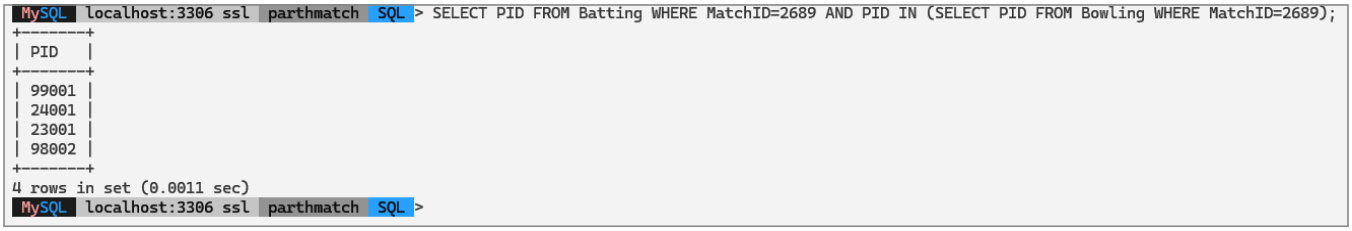
OR (Using Join)

mysql> SELECT MatchID FROM Batting,Player WHERE PID=PlayerID AND LName='Tendulkar';



Q4. Find Ids of all players that have both bowled and batted in the ODI match 2689

mysql> SELECT PID FROM Batting WHERE MatchID=2689 AND PID IN (SELECT PID FROM Bowling WHERE MatchID=2689);



Q5. Find Ids of all players that have either bowled or batted (or did both) in ODI match 2689

mysql> SELECT PID FROM Batting WHERE MatchID=2689 OR PID IN (SELECT PID FROM Bowling WHERE MatchID=2689);

```
MySQL localhost:3306 ssl parthmatch SQL > SELECT PID FROM Batting WHERE MatchID=2689 OR PID IN (SELECT PID FROM Bowling WHERE MatchID=2689);
+-------+
| PID   |
+-------+
| 23001 |
| 89001 |
| 23001 |
| 25001 |
| 99002 |
| 95001 |
| 24001 |
| 99001 |
| 27001 |
| 98002 |
| 98002 |
+-------+
11 rows in set (0.0008 sec)
MySQL localhost:3306 ssl parthmatch SQL >
```

Using Union

mysql> SELECT PID FROM Batting WHERE MatchID=2689 UNION (SELECT PID FROM Bowling WHERE MatchID=2689);

```
MySQL localhost:3306 ssl parthmatch SQL > SELECT PID FROM Batting WHERE MatchID=2689 UNION (SELECT PID FROM Bowling WHERE MatchID=2689);
+-------+
| PID   |
+-------+
| 89001 |
| 23001 |
| 25001 |
| 99002 |
| 95001 |
| 24001 |
| 99001 |
| 27001 |
| 98002 |
+-------+
9 rows in set (0.0009 sec)
MySQL localhost:3306 ssl parthmatch SQL >
```

Q6. Find Ids

mysql> SELECT PID FROM Batting WHERE MatchID=2689 AND PID NOT IN (SELECT PID FROM Bowling WHERE MatchID=2689);

```
MySQL localhost:3306 ssl parthmatch SQL > SELECT PID FROM Batting WHERE MatchID=2689 AND PID NOT IN (SELECT PID FROM Bowling WHERE MatchID=2689);
+-------+
| PID   |
+-------+
| 89001 |
| 25001 |
| 99002 |
| 95001 |
| 27001 |
+-------+
5 rows in set (0.0011 sec)
MySQL localhost:3306 ssl parthmatch SQL >
```

Q7. Find match Ids where Tendulkar has played.

mysql> SELECT MatchID FROM Batting,(SELECT PlayerID FROM Player WHERE LName='Tendulkar') ST WHERE PID=ST.PlayerID;

```
MySQL localhost:3306 ssl parthmatch SQL > SELECT MatchID FROM Batting,(SELECT PlayerID FROM Player WHERE LName='Tendulkar') ST WHERE PID=ST.PlayerID;
+---------+
| MatchID |
+---------+
|    2689 |
+---------+
1 row in set (0.0010 sec)
MySQL localhost:3306 ssl parthmatch SQL >
```

Q8(a). Find match Ids of all those matches in which player 27001 bats and makes more runs than he made at every match he played at Colombo

mysql> SELECT MatchID FROM Batting WHERE PID=27001 AND NRuns > ALL (SELECT NRuns FROM Batting WHERE PID=27001 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Colombo'));

```
MySQL localhost:3306 ssl parthmatch SQL > SELECT MatchID FROM Batting WHERE PID=27001 AND NRuns > ALL (SELECT NRuns FROM Batting WHERE PID=27001 AND MatchID IN (SELECT  MatchID FROM ODIMatch WHERE Ground='
Colombo'));
+---------+
| MatchID |
+---------+
|    2689 |
+---------+
1 row in set (0.0012 sec)
MySQL localhost:3306 ssl parthmatch SQL >
```

Q8(b). Find match Ids of all those matches in which player 27001 has scored minimum runs than he made at every match he played at Brisbane

mysql> SELECT MatchID FROM Batting WHERE PID=27001 AND NRuns < ALL (SELECT NRuns FROM Batting WHERE PID=27001 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Brisbane'));

```
MySQL localhost:3306 ssl parthmatch SQL > SELECT MatchID FROM Batting WHERE PID=27001 AND NRuns < ALL (SELECT NRuns FROM Batting WHERE PID=27001 AND MatchID IN (SELECT  MatchID FROM ODIMatch WHERE Ground=
Brisbane'));
+---------+
| MatchID |
+---------+
|    2755 |
+---------+
1 row in set (0.0009 sec)
MySQL localhost:3306 ssl parthmatch SQL >
```

Q9(a). Find Ids and scores of players who scored less than 75 but more than 50 in Colombo.

mysql> SELECT PID,NRuns FROM Batting WHERE NRuns BETWEEN 51 AND 74 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Colombo');

```
MySQL localhost:3306 ssl parthmatch SQL > SELECT PID,NRuns FROM Batting WHERE NRuns BETWEEN 51 AND 74 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE  Ground='Colombo');
+-------+-------+
| PID   | NRuns |
+-------+-------+
| 25001 |    71 |
| 91001 |    60 |
+-------+-------+
2 rows in set (0.0008 sec)
MySQL localhost:3306 ssl parthmatch SQL >
```

Q9(b). Find Ids and scores of players who scored greater than 49 runs in Colombo.

mysql> SELECT PID,NRuns FROM Batting WHERE NRuns > 49 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Colombo');

```
MySQL localhost:3306 ssl parthmatch SQL > SELECT PID,NRuns FROM Batting WHERE NRuns > 49 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Colombo');
+-------+-------+
| PID   | NRuns |
+-------+-------+
| 25001 |    71 |
| 91001 |    60 |
+-------+-------+
2 rows in set (0.0009 sec)
MySQL localhost:3306 ssl parthmatch SQL >
```

Q10. Find the player Ids of those players whose date of first match (FTest) is not given in the database.

mysql> SELECT PlayerID FROM Player WHERE FTest IS NULL;

```
MySQL localhost:3306 ssl parthmatch SQL > SELECT PlayerID FROM Player WHERE FTest IS NULL;
+----------+
| PlayerID |
+----------+
|    27001 |
+----------+
1 row in set (0.0007 sec)
```

Q11. Find names of the player who batted in match 2689

mysql> SELECT FName,LName FROM Player WHERE EXISTS (SELECT * FROM Batting WHERE PlayerID=PID AND MatchID=2689);

OR

mysql> SELECT FName,LName FROM Player,Batting WHERE PlayerID = PID AND MatchID=2689;

```
MySQL localhost:3306 ssl parthmatch SQL >  SELECT FName,LName FROM Player WHERE EXISTS ( SELECT * FROM Batting WHERE PlayerID=PID AND MatchID=2689);
+-----------+-----------+
| FName     | LName     |
+-----------+-----------+
| Sachin    | Tendulkar |
| Yuvraj    | Singh     |
| MS        | Dhoni     |
| Adam      | Gilchrist |
| Ricky     | Ponting   |
| Andrew    | Symonds   |
| Breet     | Lee       |
| Praveen   | Kumar     |
| Harbhajan | Singh     |
+-----------+-----------+
9 rows in set (0.0009 sec)
MySQL localhost:3306 ssl parthmatch SQL > SELECT FName,LName FROM Player,Batting WHERE PlayerID = PID AND MatchID=2689;
+-----------+-----------+
| FName     | LName     |
+-----------+-----------+
| Sachin    | Tendulkar |
| Yuvraj    | Singh     |
| MS        | Dhoni     |
| Adam      | Gilchrist |
| Ricky     | Ponting   |
| Andrew    | Symonds   |
| Breet     | Lee       |
| Praveen   | Kumar     |
| Harbhajan | Singh     |
+-----------+-----------+
9 rows in set (0.0008 sec)
MySQL localhost:3306 ssl parthmatch SQL >
```

Q12. Find player Ids of player who have scored more than 30 in every ODI match that they have played.

mysql> SELECT PID AS PlayerID FROM Batting b1 WHERE NOT EXISTS (SELECT * FROM Batting b2 WHERE b1.PID=b2.PID AND NRuns < 31);

```
MySQL localhost:3306 ssl parthmatch SQL > SELECT PID AS PlayerID FROM Batting b1 WHERE NOT EXISTS (SELECT * FROM Batting b2 WHERE b1.PID=b2.PID AND NRuns < 31);
+----------+
| PlayerID |
+----------+
|    24001 |
|    25001 |
|    25001 |
|    89001 |
|    91001 |
+----------+
5 rows in set (0.0014 sec)
MySQL localhost:3306 ssl parthmatch SQL >
```

Q13. Find the names of players that have batted in all the ODI matches in Melbourne that are in the database.

```
SELECT FName, LName
FROM Player
WHERE NOT EXISTS
(SELECT *
FROM ODIMatch
WHERE Ground = 'Melbourne'
AND NOT EXISTS
(SELECT *
FROM Batting
WHERE PID=PlayerID
AND (Batting.MatchId=ODIMatch.MatchID)));
```

```
MySQL localhost:3306 ssl parthmatch SQL > SELECT FName, LName FROM Player WHERE NOT EXISTS (SELECT * FROM ODIMatch WHERE Ground='Melbourne' AND NOT EXISTS (SELECT * FROM Batting WHERE PID=PlayerID AND Batt
ing.MatchID=ODIMatch.MatchID));
Empty set (0.0029 sec)
MySQL localhost:3306 ssl parthmatch SQL >
```

Q14. Find the number of players that bowled in ODI match 2689

mysql> SELECT COUNT(*) AS NBowlers FROM Bowling WHERE MatchID=2689;

```
MySQL localhost:3306 ssl parthmatch SQL > SELECT COUNT(*) AS NBowlers FROM Bowling WHERE MatchID=2689;
+----------+
| NBowlers |
+----------+
|        4 |
+----------+
1 row in set (0.0009 sec)
MySQL localhost:3306 ssl parthmatch SQL >
```

Q15. Find the average batting score of all the players that batted in the ODI Match 2689

mysql> SELECT AVG(NRuns) AS AvgRuns_2689 FROM Batting WHERE MatchID=2689;

```
MySQL localhost:3306 ssl parthmatch SQL > SELECT AVG(NRuns) AS AvgRuns_2689 FROM Batting WHERE MatchID=2689;
+--------------+
| AvgRuns_2689 |
+--------------+
|      25.2222 |
+--------------+
1 row in set (0.0007 sec)
MySQL localhost:3306 ssl parthmatch SQL >
```

Q16. Find the youngest player in the database

mysql> SELECT FName AS Youngest_Player FROM Player WHERE YBorn=(SELECT MAX(YBorn) FROM Player);

```
MySQL localhost:3306 ssl parthmatch SQL >  SELECT FName AS Youngest_Player FROM Player WHERE YBorn=(SELECT MAX(YBorn) FROM Player);
+-----------------+
| Youngest_Player |
+-----------------+
| Ishant          |
+-----------------+
1 row in set (0.0029 sec)
MySQL localhost:3306 ssl parthmatch SQL >
```

Q17. Find the number of players in the database from each country

mysql> SELECT Country, COUNT(*) AS NPlayers FROM Player GROUP BY Country;

```
MySQL localhost:3306 ssl parthmatch SQL > SELECT Country, COUNT(*) AS NPlayers FROM Player GROUP BY Country;
+--------------+----------+
| Country      | NPlayers |
+--------------+----------+
| India        |       10 |
| Australia    |        5 |
| West Indies  |        1 |
| Sri Lanka    |        3 |
| Pakistan     |        2 |
| Zimbabwe     |        1 |
| South Africa |        4 |
| New Zealand  |        2 |
| England      |        1 |
+--------------+----------+
9 rows in set (0.0008 sec)
MySQL localhost:3306 ssl parthmatch SQL >
```

Q18. Find batting average of each player in the database. Present the PID of each Player.

mysql> SELECT PID,AVG(NRuns) AS AVGR FROM Batting GROUP BY PID;

```
MySQL localhost:3306 ssl parthmatch SQL > SELECT PID,AVG(NRuns) AS AVGR FROM Batting GROUP BY PID;
+-------+---------+
| PID   | AVGR    |
+-------+---------+
| 23001 | 19.0000 |
| 24001 | 42.0000 |
| 25001 | 53.5000 |
| 27001 |  4.5000 |
| 89001 | 91.0000 |
| 91001 | 60.0000 |
| 92002 |  1.0000 |
| 94002 | 17.0000 |
| 95001 |  1.0000 |
| 98002 |  2.5000 |
| 99001 |  7.0000 |
| 99002 |  2.0000 |
+-------+---------+
12 rows in set (0.0007 sec)
MySQL localhost:3306 ssl parthmatch SQL >
```

Solve the following

Q19. Find batting average of each player from India in the database. Present first name of each player.

```
MySQL localhost:3306 ssl parthmatch SQL > select playerID,Fname,Lname,AVG(NRuns) from player JOIN batting ON player.playerID = batting.PID Group by playerID;
+----------+-----------+-------------+------------+
| playerID | Fname     | Lname       | AVG(NRuns) |
+----------+-----------+-------------+------------+
|    23001 | Yuvraj    | Singh       |    19.0000 |
|    25001 | MS        | Dhoni       |    53.5000 |
|    91001 | Sanath    | Jayasuriya  |    60.0000 |
|    94002 | Chaminda  | Vaas        |    17.0000 |
|    92002 | Muthiah   | Murlitharan |     1.0000 |
|    89001 | Sachin    | Tendulkar   |    91.0000 |
|    99002 | Adam      | Gilchrist   |     2.0000 |
|    95001 | Ricky     | Ponting     |     1.0000 |
|    24001 | Andrew    | Symonds     |    42.0000 |
|    99001 | Breet     | Lee         |     7.0000 |
|    27001 | Praveen   | Kumar       |     4.5000 |
|    98002 | Harbhajan | Singh       |     2.5000 |
+----------+-----------+-------------+------------+
12 rows in set (0.0012 sec)
MySQL localhost:3306 ssl parthmatch SQL >
```

Q20. Find the average score for each player when playing against Australia

```
MySQL localhost:3306 ssl parthmatch SQL > SELECT PlayerID, FName, LName, AVG(NRuns) AS AvgRuns
                                       -> FROM Player
                                       -> JOIN Batting ON PlayerID = PID
                                       -> JOIN ODIMatch ON Batting.MatchID = ODIMatch.MatchID
                                       -> WHERE Team1 = 'Australia' OR Team2 = 'Australia'
                                       -> GROUP BY PlayerID, FName, LName;
+----------+-----------+-----------+---------+
| PlayerID | FName     | LName     | AvgRuns |
+----------+-----------+-----------+---------+
|    89001 | Sachin    | Tendulkar | 91.0000 |
|    23001 | Yuvraj    | Singh     | 38.0000 |
|    25001 | MS        | Dhoni     | 36.0000 |
|    99002 | Adam      | Gilchrist |  2.0000 |
|    95001 | Ricky     | Ponting   |  1.0000 |
|    24001 | Andrew    | Symonds   | 42.0000 |
|    99001 | Breet     | Lee       |  7.0000 |
|    27001 | Praveen   | Kumar     |  7.0000 |
|    98002 | Harbhajan | Singh     |  3.0000 |
+----------+-----------+-----------+---------+
9 rows in set (0.0011 sec)
MySQL localhost:3306 ssl parthmatch SQL >
```

Q21. Find the ID of players that had a higher average score for all players when they played against Sri Lanka.

```
9 rows in set (0.0011 sec)
MySQL localhost:3306 ssl parthmatch SQL > SELECT B.PID
                                       -> FROM Batting B
                                       -> JOIN ODIMatch M ON B.MatchID = M.MatchID
                                       -> WHERE M.Team1 = 'Sri Lanka' OR M.Team2 = 'Sri Lanka'
                                       -> GROUP BY B.PID
                                       -> HAVING AVG(B.NRuns) > (
                                       ->     SELECT AVG(B2.NRuns)
                                       ->     FROM Batting B2
                                       ->     JOIN ODIMatch M2 ON B2.MatchID = M2.MatchID
                                       ->     WHERE M2.Team1 = 'Sri Lanka' OR M2.Team2 = 'Sri Lanka'
                                       -> );
+-------+
| PID   |
+-------+
| 25001 |
| 91001 |
+-------+
2 rows in set (0.0020 sec)
MySQL localhost:3306 ssl parthmatch SQL >
```